區間估計

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DATASET OVERVIEW

ANSUR II: Female stature

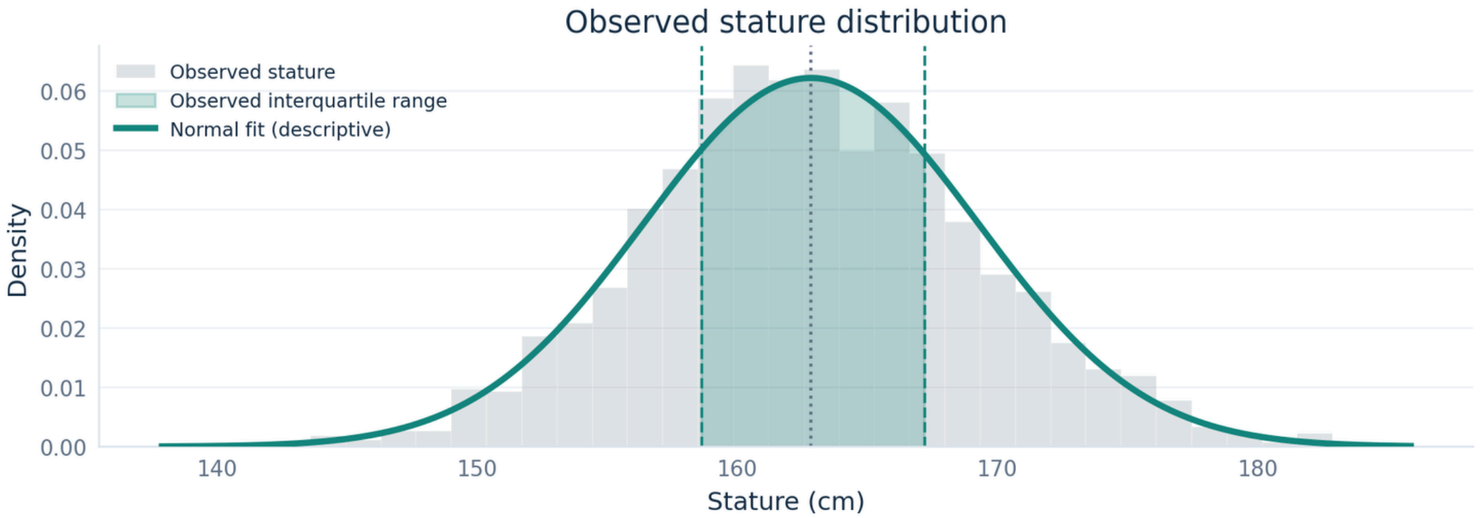
U.S. Army anthropometric survey / Observed measurements

Survey source: U.S. Army ANSUR II public data

Public files: Matthew P. Reed, University of Michigan

Variable: measured standing stature (stature), mm converted to cm

FEMALE PARTICIPANTS	MEAN STATURE	SAMPLE STANDARD DEVIATION
1,986	162.85 cm	6.42 cm



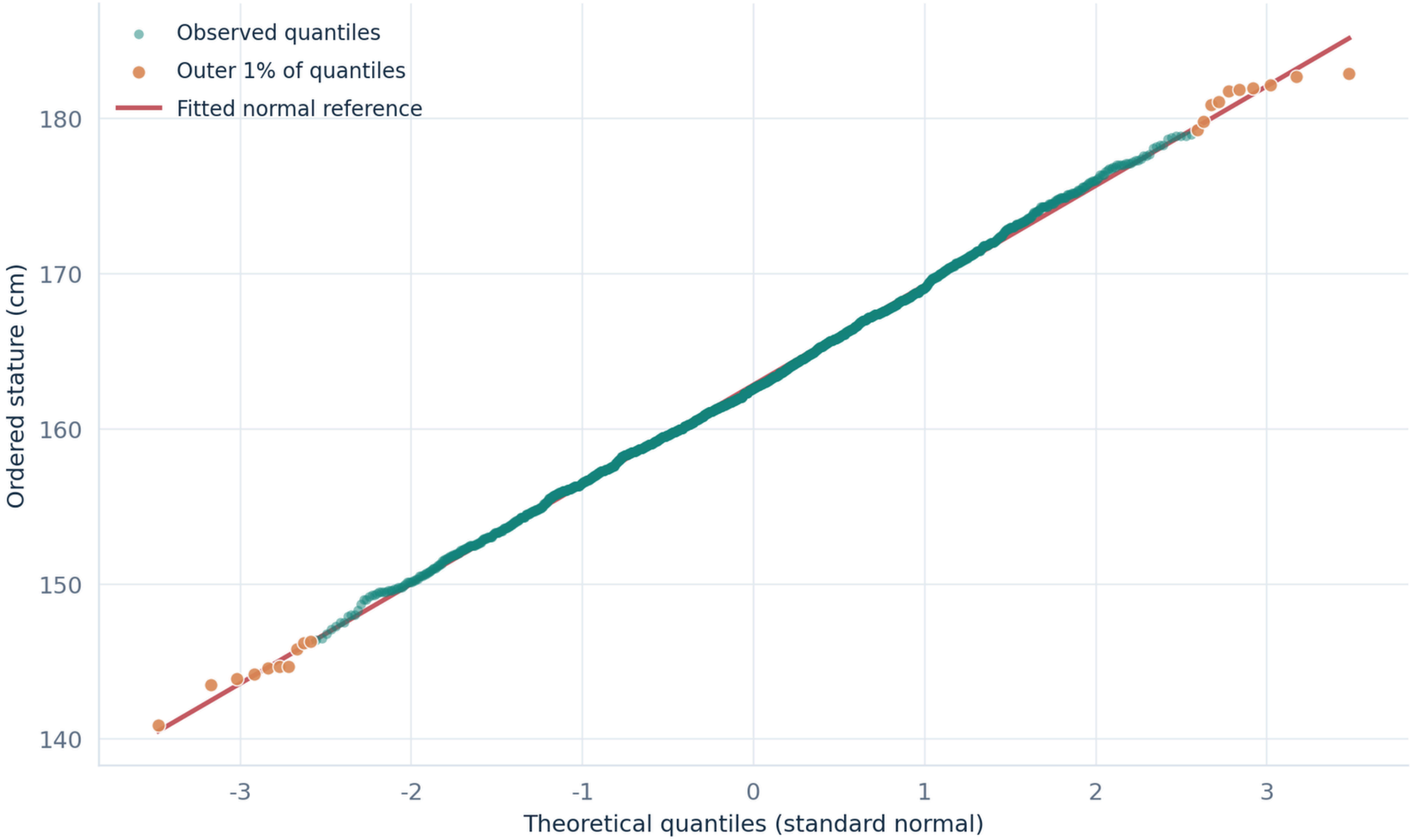
Dashed: observed quartiles. Dotted: sample mean.

Observed Army measurements; not simulated data or a civilian population estimate.

Normal Q–Q plot

ANSUR II / Female stature / Observed data

n = 1,986
Shapiro–Wilk p = 0.12971



Reference: observed mean + observed SD × theoretical normal quantile.

01 / CONFIDENCE INTERVAL

Population mean

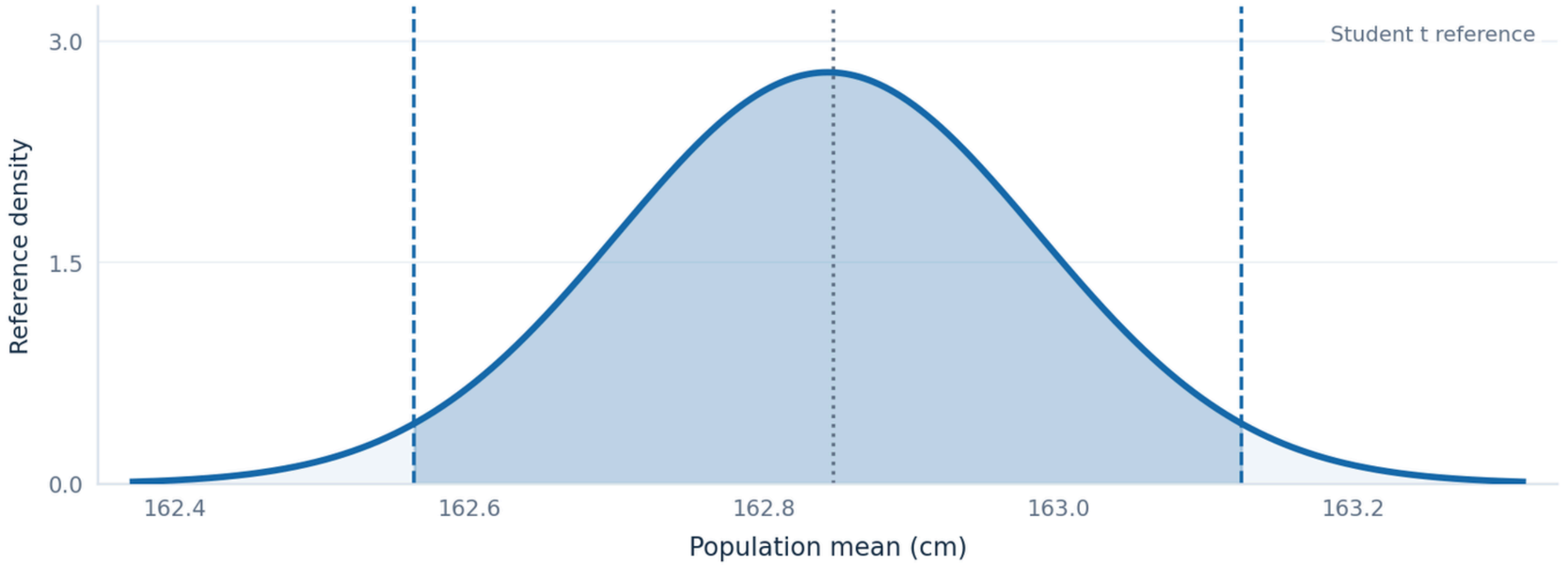
Two-sided 95% confidence interval

`Population_mean=162.84733`

(one simulation outcome)

`CI=[162.56283 , 163.12392]`

`Coverage prob.=0.9496`



Dashed: interval limits. Dotted: population reference.

CI construction reference; not a parameter posterior.

02 / CONFIDENCE INTERVAL

Population standard deviation

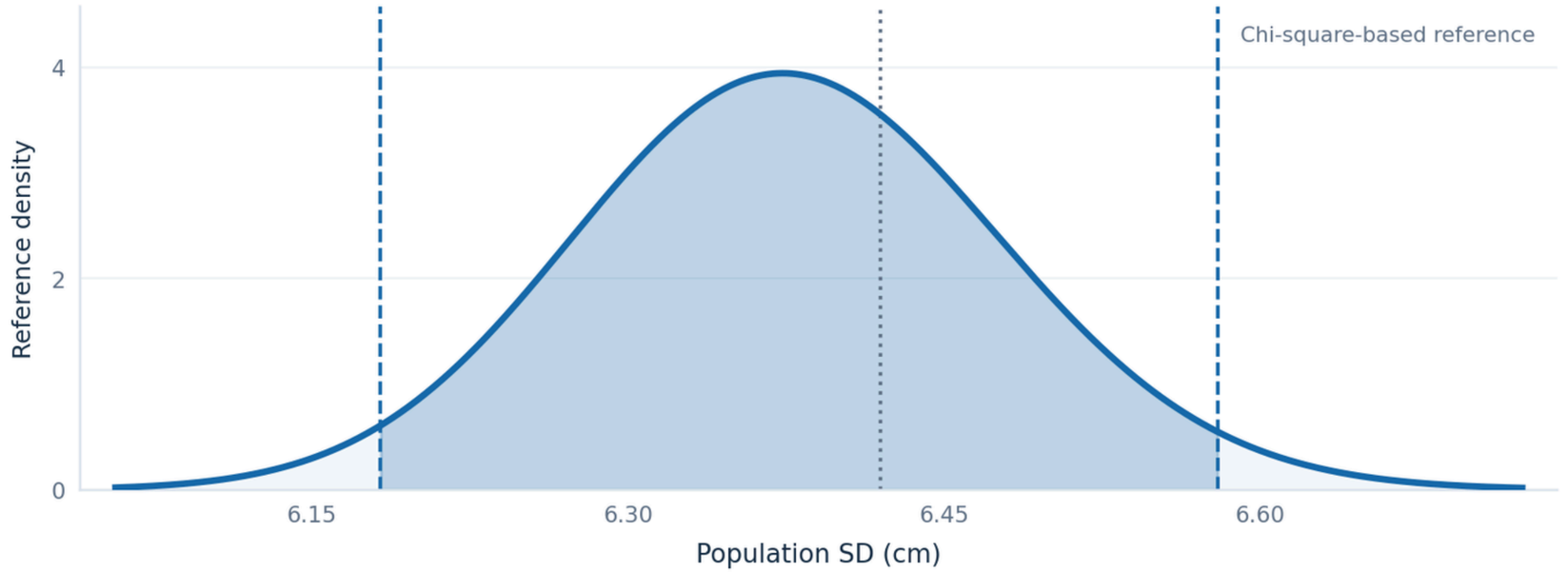
Two-sided 95% confidence interval

`Population_sd=6.41995`

(one simulation outcome)

`CI=[6.18277 , 6.57973]`

`Coverage prob.=0.9507`



Dashed: interval limits. Dotted: population reference.

CI construction reference; not a parameter posterior.

03 / PREDICTION INTERVAL

All 10 future observations

Two-sided 95% simultaneous prediction interval

(one simulation outcome)

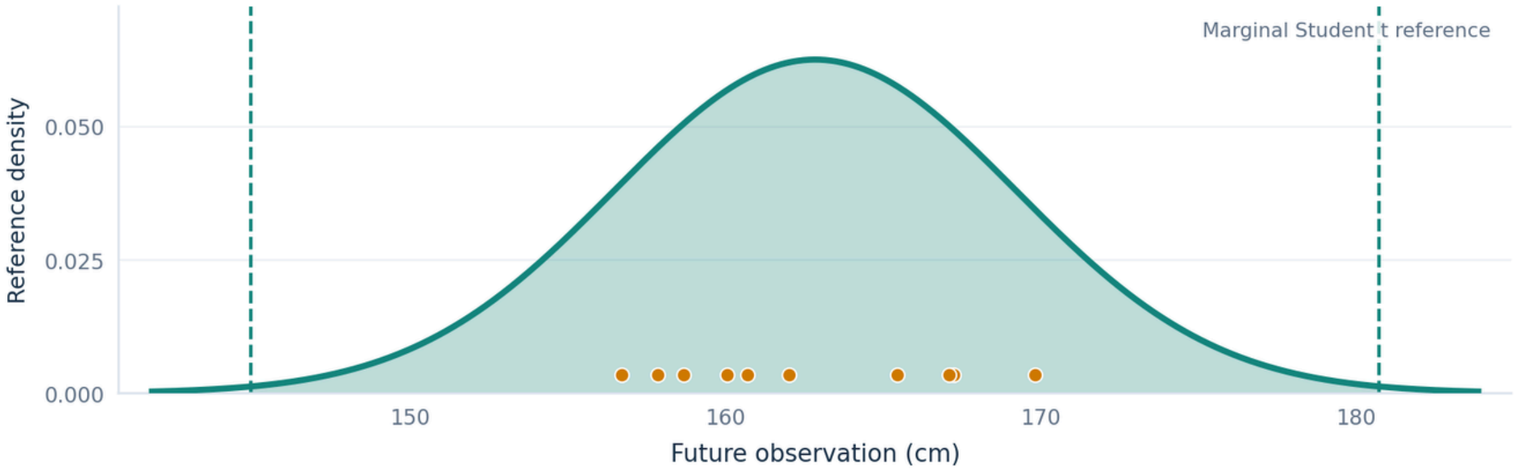
Future observations (cm), m=10:

[156.71085, 162.02608, 167.24123, 160.05907, 167.09708,
169.80182, 165.44962, 157.86089, 160.68846, 158.67769]

PI=[144.97219 , 180.71455]

All 10 inside PI: Yes (10/10)

Coverage prob.=0.9506



Dashed: interval limits. Orange dots: 10 simulated future observations.

Marginal shading; 95% coverage applies to all 10 jointly.

04 / PREDICTION INTERVAL

Mean of 5 future observations

Two-sided 95% prediction interval

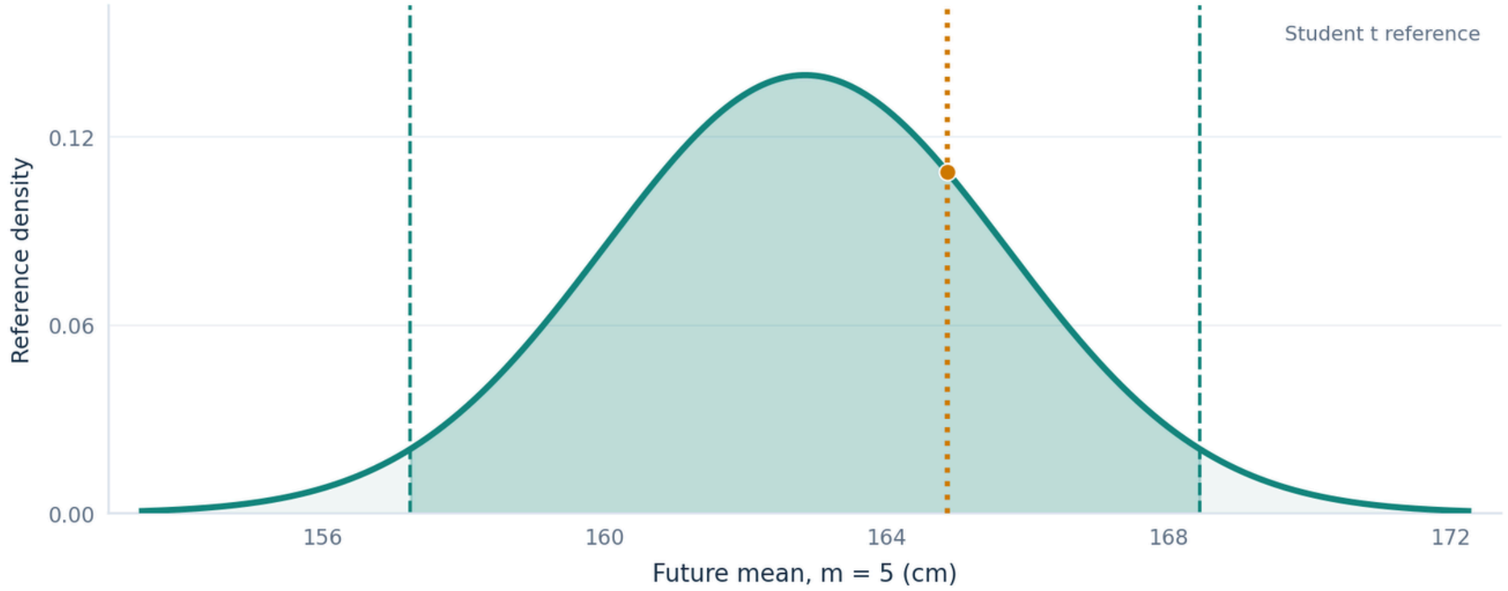
`Future_mean (m=5)=164.86499`

(one simulation outcome)

`PI=[157.24507 , 168.44168]`

`Future mean inside PI: Yes`

`Coverage prob.=0.9508`



Dashed: interval limits. Orange: simulated future mean.

Shading: 95% reference range for the future statistic.

05 / PREDICTION INTERVAL

SD of 5 future observations

Two-sided 95% prediction interval

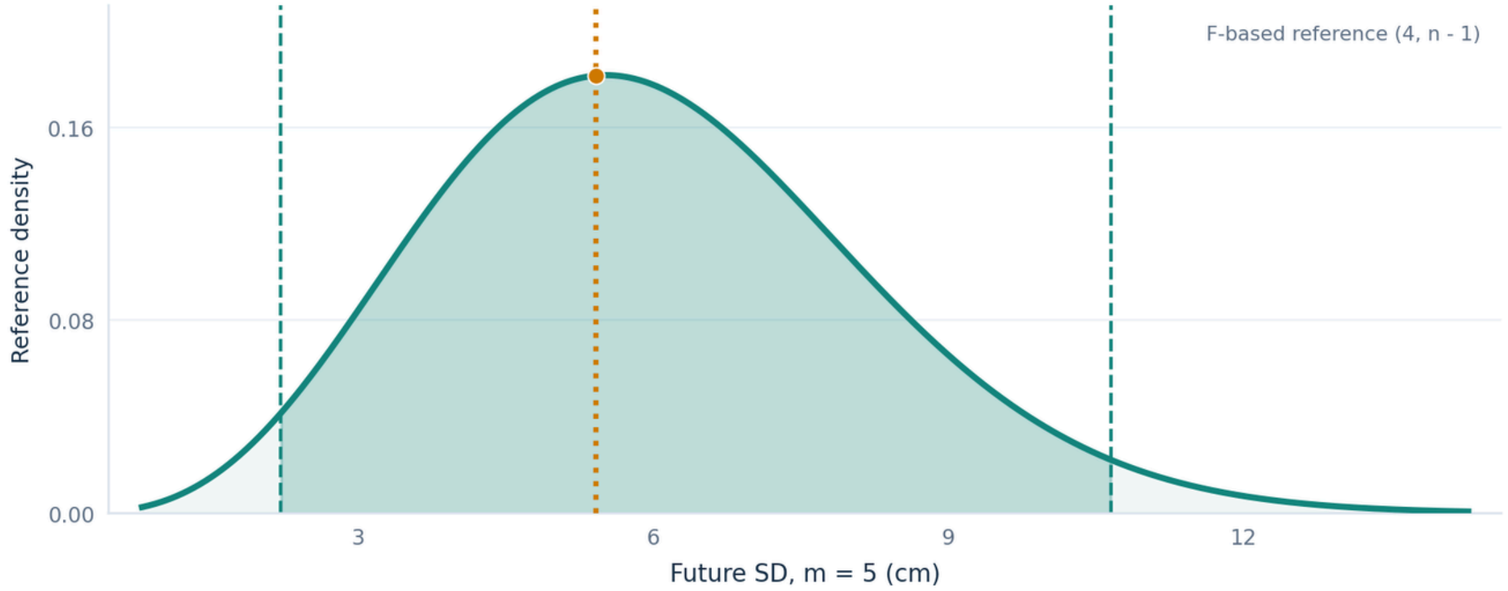
`Future_sd (m=5, ddof=1)=5.42143`

(one simulation outcome)

`PI=[2.21809 , 10.65269]`

`Future SD inside PI: Yes`

`Coverage prob.=0.9497`



Dashed: interval limits. Orange: simulated future sample SD.

Shading: 95% reference range for the future statistic.

06 / TOLERANCE INTERVAL

At least 99% of the population

Two-sided tolerance interval with 95% confidence

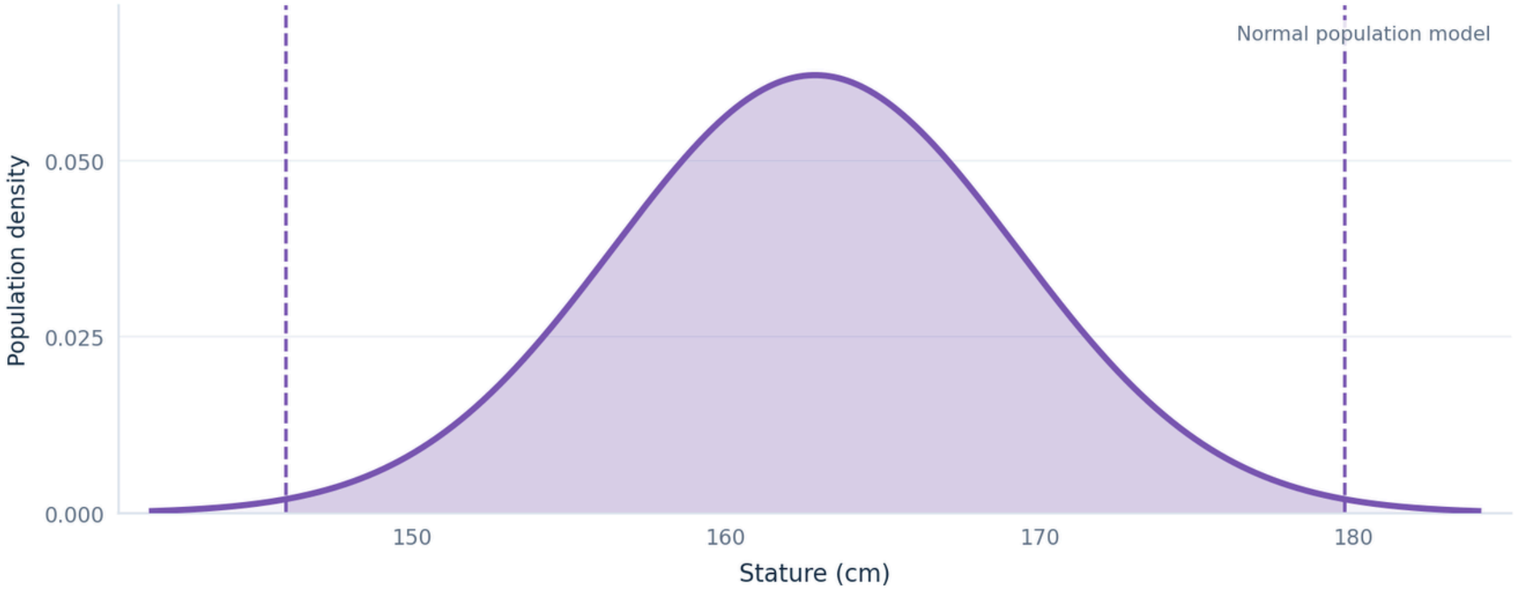
Population coverage=99.13907%

(one simulation outcome)

TI=[145.97694 , 179.70980]

At least 99% covered: Yes

Coverage prob.=0.9502



Dashed: interval limits. Shading: modeled population content.

Shading: population content. Confidence level: 95%.

Conclusion

- 六種區間中有4種的模擬涵蓋率高於0.95

- 未來10筆觀測的PI最寬，而母體標準差的CI最窄



Thanks!